

# Neuro-Symbolic Agents Playing Magic: The Gathering

Ontology-Grounded Knowledge Graphs, Latent World Models,  
and Control-Theoretic Action Selection under Partial Observability

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## Abstract

*Magic: The Gathering* (MTG) is, by several measures, the most combinatorially complex commercial card game ever published: more than 28,000 unique cards, a 260-page rule book, partial observability over 50–90 hidden cards, an unbounded action space, a turn-based stack with priority, and a state space conservatively estimated above  $10^{800}$ . We study agents that play it *neuro-symbolically*: a verified rules engine decides what is legal, a formal ontology fixes what card semantics mean, a knowledge graph accumulates strategic structure, and learned components supply the statistics that no amount of symbolic knowledge can provide. Three claims organise the work. First, the symbolic layer should be a real ontology rather than a schema: we develop a provenance-aware OWL ontology of MTG aligned to DOLCE, with OWL 2 DL as its modelling target and its actual qualification boundaries described separately (Neubürger and Neubürger, 2026). Second, the knowledge graph should be *built by playing*: agents mine their own trajectories into an append-only, provenance-bearing extension layer that later agents query as a strategic prior, while curated card semantics remain untouched — a population-level learning loop in which the graph is shared long-term memory rather than static context. Third, action selection should be posed as *stochastic optimal control under partial observability* rather than as expected-free-energy minimisation: following Kaufmann (2026), the canonical discretisation of Active Inference is belief-space KL control, so we implement that objective directly and retain the two conventional expected-free-energy factorisations only as ablation arms with documented failure modes. We describe the architecture, state where the symbolic/sub-symbolic boundary lies and what crosses it in each direction, and specify the evaluation that would test the control claim inside MTG rather than on toy POMDPs. Code, ontology, and traces are released under the MIT licence.

**Keywords:** neuro-symbolic AI, knowledge graphs, world models, JEPA, stochastic optimal control, partially observable Markov decision processes, LLM agents, self-play, Magic: The Gathering.

## 1 Introduction

Reinforcement-learning (RL) research on games has historically progressed through a sequence of increasingly hard environments: backgammon (Tesauro, 1995), chess and Go (Silver et al., 2017, 2018), StarCraft II (Vinyals et al., 2019), and no-limit Texas hold’em poker (Brown and Sandholm, 2018). Each step pushed the community to invent new tools — temporal-difference (TD) learning, deep value networks, self-play, counterfactual regret minimisation (CFR). *Magic: The Gathering* (MTG) sits beyond the union of these challenges (Table 1): its action space is unbounded, its rules are open-textured natural language printed on cards (the comprehensive

Table 1: Approximate combinatorial complexity comparison.

|                        | Chess             | Go               | No-limit Poker (HU) | MTG                           |
|------------------------|-------------------|------------------|---------------------|-------------------------------|
| State space            | $10^{47}$         | $10^{170}$       | $10^{160}$          | $> 10^{800}$                  |
| Branching factor       | $\sim 35$         | $\sim 250$       | $\sim 10$           | 0–100+ (unbounded)            |
| Hidden information     | none              | none             | 2 cards             | 50–90+ cards                  |
| Card / piece pool      | 6 types           | 1 type           | 52 cards            | $> 28,000$ unique             |
| Rule text length       | $\sim 10\text{p}$ | $\sim 5\text{p}$ | $\sim 10\text{p}$   | $\sim 260\text{p}$ + per-card |
| Rules update frequency | decades           | decades          | decades             | quarterly                     |

rule book itself defers to the per-card text in many places), its hidden information includes the entire opponent’s deck composition, and its mechanics are extended every quarter by the publisher Wizards of the Coast. A single MTG card can override the rules of the game (e.g. *Yixlid Jailer*, *Blood Moon*, *Cumulative Upkeep*), and a single decision in combat may resolve dozens of triggered abilities in a player-chosen order. The state space is open in the strict sense: every quarterly set adds previously-unseen card text whose semantics must be parsed at runtime, so any agent that hard-codes its action vocabulary cannot survive a set rotation.

For a concrete sense of why this matters for an algorithmic agent: in a single midgame turn the active player may need to (i) decide whether to play a land, (ii) sequence two or three spells with mana cost, colour requirements and stack ordering, (iii) optionally activate any number of permanent abilities that themselves go on the stack, (iv) declare attackers under summoning-sickness and tapping rules, (v) respond to triggered abilities resolving in the order chosen by their controllers, and (vi) decide which damage to assign in combat under multi-blocker rules. Each of these is itself a sub-decision tree. The total number of legal actions per priority pass routinely exceeds 60 and, in pathological combo states, several hundred. A model-free policy that emits a fixed-size logit vector cannot represent this branching, and a tabular search procedure that enumerates the subtree cannot fit it into memory.

A purely model-free deep-RL approach is therefore unattractive: the action space cannot be enumerated as a fixed-size logit vector, the observation space is symbolic and irregular, and every new set introduces previously-unseen card text whose semantics must be parsed at run time. Conversely, a purely symbolic LLM-prompting approach struggles with rules-correctness, hallucinated cards, and the long-horizon credit-assignment that competitive play requires — an LLM that simply names a move it would like to make has no guarantee that move is legal, that it resolves the way the LLM expects, or that it furthers a coherent multi-turn plan.

Our position is that the structure of MTG demands a *neuro-symbolic* stack: a verified rules engine to keep play legal, a knowledge graph to make card semantics queryable, a learned world model to enable imagination and planning over the resulting compressed state, and an LLM agent that orchestrates these components under an active-inference objective. Figure 1 sketches the resulting architecture as a data-flow diagram, separating the *symbolic* substrate (data sources, engine, knowledge graph) from the *learned* substrate (world model, agent zoo) and showing where information actually crosses the boundary.

**Contributions.** This work makes four contributions.

1. A scientific framing of MTG as an *open-ended partially observable control problem* whose difficulty comes not only from combinatorics, but from the continual introduction of new symbolic rules through quarterly card releases.
2. An architecture in which the symbolic/sub-symbolic boundary is stated rather than assumed

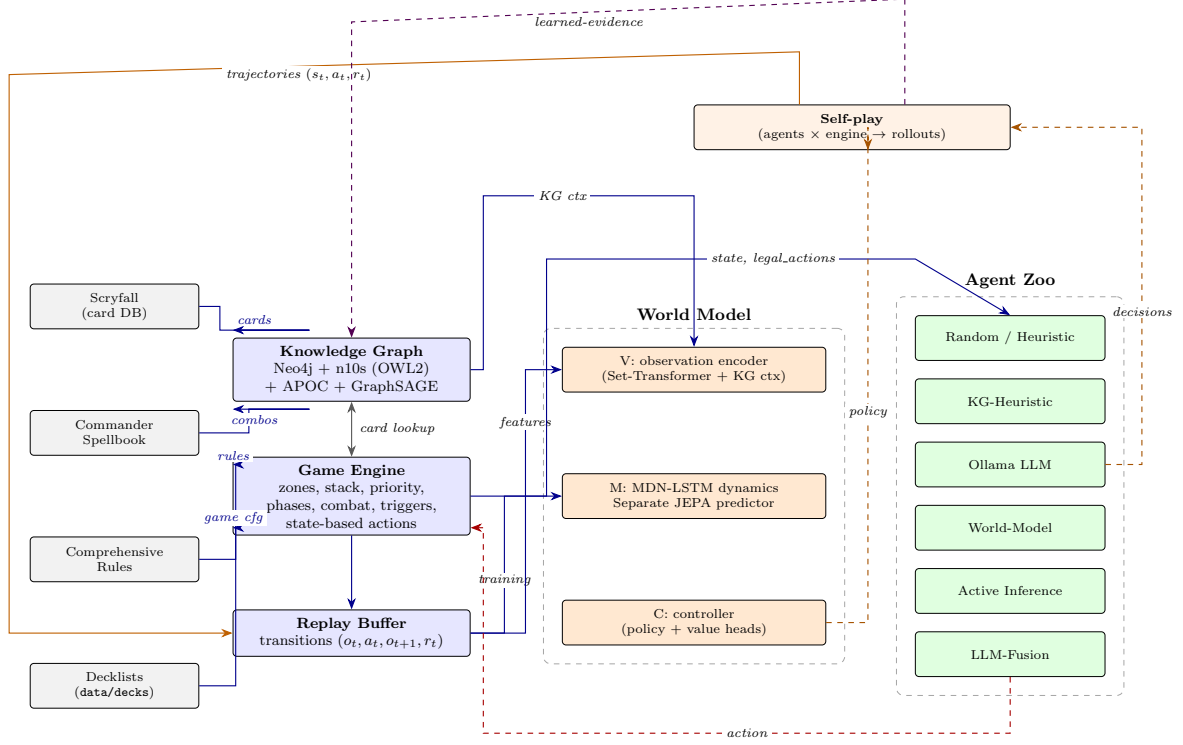


Figure 1: Data-flow architecture. **Grey** boxes are external data sources; **blue** the symbolic substrate (engine, knowledge graph, replay buffer); **orange** the learned world model (V+M+C with a JEPA target head and a knowledge-graph context channel); **green** the agent zoo. Solid blue arrows are forward data flow; the red dashed arrow is the agent’s chosen action returned to the engine; the engine  $\leftrightarrow$  knowledge-graph link is bidirectional (card lookup at decision time, ontology-grounded card metadata at engine boot). Each agent additionally has read-only access to the knowledge graph and to the world-model controller through a thin lookup interface (omitted from the figure for clarity, described in the body text). The engine is the only component allowed to mutate game state; agents read but do not write.

(Section 3.5): what is symbolic, what is learned, what crosses in each direction, and what gates the crossing.

3. A *play-to-graph* learning loop in which agents mine their own trajectories into an append-only extension layer of the knowledge graph, and later agents query that layer as a strategic prior — with curated card semantics protected from statistical contamination by construction (Section 4.3).
4. A *control-theoretic* reading of MTG decision making: we adopt the result of Kaufmann (2026) that discrete Active Inference is belief-space KL control, implement that objective directly, and retain the two legacy expected-free-energy factorisations only as ablation arms (Section 3.3).

The ontology that grounds the symbolic layer is a contribution in its own right and is described in a companion paper (Neubürger and Neubürger, 2026); here we treat it as given and focus on what an agent does with it.

**Why these four together.** Each contribution targets a specific failure mode of the others. The engine prevents the LLM from playing illegal moves and the world model from imagining inconsistent dynamics. The knowledge graph compresses the 28,000-card pool into a structure with a few hundred archetype clusters so that an agent can reason about “what card type does

this state want next” without enumerating all candidates. The world model amortises the cost of multi-turn lookahead so that decisions remain well within a few hundred milliseconds even when planning ten turns deep. The active-inference fusion agent finally ties the three together with a single tunable objective — expected free energy — whose pragmatic and epistemic terms have a direct mapping onto MTG concepts of *value* and *information*.

**Collective intelligence hypothesis.** Beyond single-agent optimisation, the same stack naturally supports a population-level learning loop: many heterogeneous agents generate trajectories, those trajectories are mined for repeated card-pair and outcome regularities, and the resulting evidence is written into an append-only graph extension layer with provenance. New agents then query this enriched graph and inherit strategic priors discovered by previous runs. In this reading, the KG is not only static symbolic context but a shared long-term memory for the entire agent population.

**Scope and what we do *not* do.** We do not solve MTG; we do not even target tournament-strength play. The contribution is a research substrate that is rules-correct on the implemented card pool, that maintains a queryable symbolic representation of card semantics, and that supports plug-in research on world models and inference-style agents on a problem qualitatively harder than the standard benchmarks. The empirical claims in §6 are therefore framework-level (decks complete, agents converge faster than baselines, ablations show graded contributions) rather than absolute strength claims against human play.

The framework is released as open source<sup>1</sup> and the codebase is the canonical artifact accompanying this report. Section 2 positions the work against prior literature; §3 formalises the POMDP and the four learning principles (world models, FEP, knowledge-graph priors, constrained-decoding LLMs); §4 walks the architecture top-down; §6 presents the controlled study design and current native evidence; §7–8 discuss limitations and next steps.

## 2 Related Work

**World models.** Ha and Schmidhuber (2018) introduced the V+M+C decomposition — a vision-VAE, recurrent mixture-density dynamics and a controller trained in imagined trajectories. Successors include Dreamer (Hafner et al., 2020, 2023) and the Schmidhuber-line of self-referential systems (Schmidhuber, 2015; Irie et al., 2022). We retain this separation but encode structured game observations with a Set-Transformer and optional symbolic context. A separate JEPA head (LeCun, 2022; Assran et al., 2023) learns action-conditioned next-embedding prediction, avoiding the requirement to generate a complete next-state description. LeWorldModel and the modular `stable-pretraining/stable-worldmodel` ecosystem provide related methods and reusable research infrastructure (Balestriero et al., 2025; Maes et al., 2026a,b). Representation prediction is a training principle, not itself a policy or a proof of strategically sufficient abstraction.

**Active inference and stochastic optimal control.** The Free-Energy Principle of Friston (2010) and its discrete-state operationalisation (Sajid et al., 2021) propose that action selection minimise *expected free energy* (EFE), conventionally decomposed into a pragmatic (reward-seeking) and an epistemic (information-seeking) term. Kaufmann (2026) shows that this framing inverts the actual mathematical hierarchy: continuous-time Active Inference on path space *is* finite-horizon path-integral stochastic MPC in the sense of Kappen (2005), Todorov (2009) and Williams et al. (2017), and its canonical discretisation is classical KL control on belief space. The exploration–exploitation trade-off is not supplied by an added epistemic term; it is Fel’dbaum’s dual control effect (Fel’dbaum, 1960), present in stochastic control from the outset, and it is

<sup>1</sup><https://github.com/cuteredpwnda/opposition-agents-playing-mtg>

recovered as soon as plans are evaluated *closed-loop* over beliefs rather than as open-loop action sequences. We follow that result rather than the conventional decomposition, for the concrete engineering reasons set out in Section 3.3: the two extant EFE factorisations have opposing, empirically demonstrated failure modes, and both are avoidable.

**Knowledge graphs, GraphRAG, and their limits.** Neo4j with the `n10s` plugin enables direct ingestion of OWL ontologies as labeled property graphs (Neo4j Labs, n.d.); the APOC library adds graph algorithms; GraphSAGE (Hamilton et al., 2017) provides inductive node embeddings. GraphRAG (Edge et al., 2024) complements vector RAG for queries that benefit from explicit relational structure, and Liao et al. (2026) give the first component-level ablation of a GraphRAG pipeline, finding that retriever choice and graph *serialisation format* dominate the outcome — structured, code-like serialisations outperform prose renderings of the same subgraph. MTG is a textbook GraphRAG case: “find me a two-card combo using only cards in my library that survive *Counterspell*” is a multi-hop graph query, not a similarity search. Two cautions from the same literature shape our design. Diettrich et al. (2026) show that consumers of an evolving schema must discover it at runtime rather than assume it, because static pipelines fail silently under schema drift; and Romanova (2026) gives a per-relation diagnostic for whether explicit graph structure still adds signal beyond node text at all, which is the honest way to decide where graph engineering is worth the effort.

**Foundational ontologies and the symbolic layer.** The intended symbolic substrate is a provenance-aware OWL ontology of MTG aligned to DOLCE-UltraLite (Gangemi et al., 2002; Borgo et al., 2022). Its conceptual schema and tooling were developed with LLM assistance; its normative vocabulary is mechanically derived from the Comprehensive Rules without runtime LLM inference. A companion paper (Neubürger and Neubürger, 2026) separates construction responsibilities and reports logical and data-validation scopes. Constraints are expressed as testable axioms, but schema consistency is not full profile conformance of the imported documents or certification of a populated agent graph. The broader turn toward competency-question-driven, validator-gated ontology construction (jiayi et al., 2026; Ontology Engineering Group, Universidad Politécnica de Madrid, 2026; Lippolis et al., 2026; Tiwari et al., 2026) is methodological context, not evidence that those external pipelines were executed here.

**Neuro-symbolic systems and the curated/induced boundary.** The framing that organises Section 3.5 is the distinction between *curated* and *induced* semantics. Sabou (2026) argues that trustworthy neuro-symbolic AI requires explanations grounded in domain knowledge rather than saliency, and identifies provenance and continuous auditing as the missing infrastructure. Thombre et al. (2026) demonstrate that neuro-symbolic decomposition — symbolic constraints narrowing the search space, the neural component reasoning inside each narrowed sub-task — substantially outperforms monolithic prompting. Van Schependom et al. (2026) generate reasoning training data from an ontology by backward-chaining proof search, the natural route to a reasoning benchmark *for* a schema such as ours. Ehrenmüller et al. (2026) address the reconciliation of conflicting graph claims from heterogeneous sources with explicit methodological provenance, which is structurally the same problem as reconciling curated card semantics with statistics mined from imperfect self-play. Our contribution to this thread is a domain where the induced layer is generated by an agent acting in a *verified* environment, so a learned assertion can in principle be traced back to the exact game states that support it.

**MTG and other complex card games.** Earlier MTG-AI work has used hand-crafted heuristics (Cowling et al., 2012), MCTS, and small policy nets on restricted card pools (e.g., `open-mtg`). On the rules side, `Forge`, `XMage` and `Mage` provide JVM-based reference implementations, and `phase-rs` provides a Rust one. To our knowledge, no prior public system combines a full

Comprehensive-Rules engine with a learned world model and a foundational-ontology-grounded symbolic layer. [Moravčík et al. \(2017\)](#) (poker), [Vinyals et al. \(2019\)](#) (StarCraft) and [Schrittwieser et al. \(2020\)](#) (MuZero) are spiritual antecedents at the algorithmic level.

**Ontologies for games.** Game ontologies support purposes including educational metadata, game design and simulation. The companion resource ([Neubürger and Neubürger, 2026](#)) targets interoperability between normative vocabulary, card descriptions, contextual gameplay and strategic evidence. Relative coverage and modelling commitments of other MTG representations have not yet been systematically compared; no priority claim is made.

### 3 Theoretical Foundations

#### 3.1 MTG as a partially-observable Markov game

We model a game of Magic from the perspective of a single seat as a finite-horizon POMDP  $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, \Omega, R, \gamma, b_0)$  embedded inside an  $n$ -player Markov game. The full state  $s_t \in \mathcal{S}$  encompasses every public and private zone of every player — libraries, hands, the stack, mana pools, and a long tail of bookkeeping flags (summoning sickness, monarchy, the day/night state, designations, emblems, dungeon room, companion eligibility). A controlled-perspective filter  $\Omega : \mathcal{S} \rightarrow \mathcal{O}$  hides the opposing players’ hands and the order of their libraries; the agent only ever observes  $o_t = \Omega(s_t)$ . The action space  $\mathcal{A}_t = \text{LEGAL}(s_t)$  is computed afresh on every priority pass and is highly variable: typically  $|\mathcal{A}_t| \in [1, 60]$ , but can spike past several hundred in hard combo states. The transition model  $T(s' | s, a)$  is the rules engine itself, fully deterministic given a random seed and the active player’s choices. Reward  $R$  is sparse and terminal —  $\pm 1$  on the win condition — but agents may shape it during training using  $\Delta\text{life}$ ,  $\Delta\text{cards}$ ,  $\Delta\text{tempo}$  as auxiliary signals. The state space is combinatorially large: a conservative count over deck-times-zone configurations gives  $|\mathcal{S}| > 10^{800}$ , several orders of magnitude beyond Go.

The optimal value function over beliefs  $b_t \in \Delta(\mathcal{S})$  satisfies the Bellman equation

$$V^*(b) = \max_{a \in \mathcal{A}} \left[ \sum_s b(s) R(s, a) + \gamma \sum_o \Pr(o | b, a) V^*(b'_o) \right], \quad (1)$$

where  $b'_o$  is the Bayes update of  $b$  after taking  $a$  and observing  $o$ . Exact solution is intractable. Each agent in our zoo is therefore an explicit approximation that trades different axes of accuracy: tabular reactive policies (Random, Heuristic, KG-Heuristic, LLM) collapse  $b_t$  to a point estimate  $s_t \approx o_t$ ; the world-model agent learns an amortised Bellman residual in latent space; the active-inference agents maintain an explicit belief and minimise expected free energy.

#### 3.2 World models as latent imagination

The V+M+C decomposition of [Ha and Schmidhuber \(2018\)](#) separates representation learning, dynamics prediction and action selection. Our inputs are structured, perspective-filtered game observations: visible card identities and characteristics, zones, resources, turn information and public actions. The relevant choice is *predicting a compact representation of future observations versus predicting a complete next game-state description*.

**What is predicted.** An encoder  $E_\phi$  maps an observation  $o_t$  and optional symbolic context  $g_t$  into a latent vector  $z_t$ . An action-conditioned predictor  $f_\theta$  predicts the representation that the encoder would assign to the next observed position:

$$z_t = E_\phi(o_t, g_t), \quad \hat{z}_{t+1} = f_\theta(z_t, a_t), \quad z_{t+1}^{\text{target}} = \text{sg}(E_\phi(o_{t+1}, g_{t+1})). \quad (2)$$



Here  $\phi$  and  $\theta$  denote learned parameters,  $a_t$  is an encoded action, and sg means *stop gradient*: the target is treated as fixed for this loss evaluation. The current and target branches use the same encoder parameters; a separately maintained target network is not assumed. Training asks whether the consequence of an action is predictable in the learned representation, rather than demanding an exact listing of every future card, flag and hidden zone.

This is the joint-embedding predictive principle (LeCun, 2022; Assran et al., 2023; Maes et al., 2026b). It can allocate capacity to regularities useful for evaluating positions without decoding a full symbolic successor state. That is an inductive bias, not a guarantee: compression may discard a decisive interaction, and unpredictable hidden information does not become observable merely because a latent exists. The learned representation  $z_t$  is also distinct from a Bayesian belief  $b_t$  over possible hidden states.

**What the training loss compares.** The present JEPA objective uses consecutive transitions  $(o_t, a_t, o_{t+1})$ :

$$\mathcal{L}_{\text{JEPA}} = \mathbb{E}_{\mathcal{D}} \left[ \text{MSE}(\hat{z}_{t+1}, z_{t+1}^{\text{target}}) + \beta_J \text{KL}(q_{\phi}(z \mid o_t, g_t) \parallel \mathcal{N}(0, I)) \right]. \quad (3)$$

The replay data  $\mathcal{D}$  contains recorded transitions. The first term measures next-embedding prediction error. The second regularises the encoder’s Gaussian distribution  $q_{\phi} = \mathcal{N}(\mu_{\phi}, \text{diag } \sigma_{\phi}^2)$  against a standard normal reference;  $\beta_J$  controls its weight. This is *not* a KL divergence against an LSTM-predicted next-state prior. Nor is it an  $H$ -step training loss: repeated rollouts are a planning operation, while this objective supervises one step at a time. Gaussian regularisation and stop-gradient define the current training choice; neither alone establishes that representations cannot collapse or that they retain all strategically relevant information.

**How predictions become decisions.** Planning starts with a candidate legal action and repeatedly applies a latent transition model to imagine a continuation. Predicted rewards or values score those continuations; their scores are aggregated to compare candidate first actions. A generic predicted-return estimate is

$$\hat{J}_H(a) = \frac{1}{K} \sum_{k=1}^K \sum_{\ell=1}^H \gamma^{\ell-1} \hat{r}_{t+\ell}^{(k)} \quad \text{with first action } a_t = a. \quad (4)$$

Here  $H$  is planning depth,  $K$  the number of sampled continuations,  $\gamma$  a discount factor and  $\hat{r}$  a predicted reward. Larger values are preferred. No complete successor state is generated and the rules engine is not called inside these imagined continuations. The real first action must still come from the engine’s legal set.

JEPA prediction, reward/value learning and search are separate components. The present dream-search path uses recurrent stochastic dynamics and a predicted-reward head; training the JEPA head does not by itself replace that rollout model. A future JEPA-driven planner must qualify this connection explicitly. Multi-step error, latent collapse, reward accuracy and the legality of imagined continuations therefore remain distinct evaluation questions; a low one-step embedding loss is not evidence of accurate long-horizon play.

### 3.3 Active inference as path-integral control

We first distinguish *belief estimation* from *action selection*. In the notation of Section 3,  $s$  is the hidden game state and  $o$  an observation. A generative model  $p_{\text{gen}}(o, s)$  describes how states produce observations; an approximate distribution  $q_{\text{bel}}(s)$  represents the agent’s current belief. Friston’s variational free energy (Friston, 2010) is

$$F[q_{\text{bel}}] = \text{KL}(q_{\text{bel}}(s) \parallel p_{\text{gen}}(s \mid o)) - \log p_{\text{gen}}(o). \quad (5)$$

For a fixed observation the evidence term is constant, so minimising  $F$  makes the approximate belief closer to the model’s posterior. It need not make that posterior less uncertain. This estimation problem does not yet specify which action to take.

**Goals, defaults and controlled predictions.** Let  $\eta$  denote the game-state coordinates on which goals are evaluated, such as survival, opponent elimination and material advantage. These are properties of a possible game position, not the learned JEPA embedding. A preference distribution  $p_{\text{pref}}(\eta)$  specifies which positions are desirable, with cost

$$V(\eta) := -\log p_{\text{pref}}(\eta). \quad (6)$$

Lower  $V$  means a more preferred position. One may specify costs directly and read  $p_{\text{pref}} \propto e^{-V}$ , up to normalisation; the costs are not calibrated win probabilities.

A policy  $\pi$  chooses actions from beliefs. Let  $\tau$  denote a horizon-length trajectory, including states, actions and observations.  $Q^\pi(\tau)$  is its predicted distribution when following  $\pi$ , accounting for hidden-state uncertainty and future observations. A default policy  $q_0(a \mid b)$  instead induces the reference trajectory distribution  $P_0(\tau)$ . This *passive prior* is a baseline way of acting, not the agent’s belief about hidden cards; “passive” does not mean that no actions occur.

**What a Gibbs tilt means.** Following Kaufmann (2026), define a target trajectory distribution by reweighting the default trajectories:

$$P_V(\tau) = \frac{1}{Z} P_0(\tau) e^{-V(\eta_T)}, \quad Z = \mathbb{E}_{P_0}[e^{-V(\eta_T)}]. \quad (7)$$

This is a *Gibbs tilt*: multiply each baseline probability by an exponential preference weight, then renormalise so probabilities sum to one. With equally probable trajectories and terminal costs 0 and 1, their weights become 1 and  $e^{-1}$ , giving approximately 0.73 and 0.27. It changes the distribution we want to approach; it does not change the rules of the game.

The control objective measures how far the policy’s predicted trajectories are from that target:

$$G(\pi) = \text{KL}(Q^\pi(\tau) \parallel P_V(\tau)). \quad (8)$$

Substituting (7) gives an interpretable decomposition:

$$G(\pi) = \underbrace{\text{KL}(Q^\pi \parallel P_0)}_{\text{deviation from default}} + \underbrace{\mathbb{E}_{Q^\pi}[V(\eta_T)]}_{\text{expected terminal cost}} + \log Z. \quad (9)$$

The constant  $\log Z$  can be dropped when comparing policies. The first term discourages costly deviations from a specified reference; the second favours good game outcomes. If the controlled and reference processes share transition/observation kernels, their action-conditioned trajectory KL decomposes into expected action-policy divergences,  $\sum_t \mathbb{E}[\text{KL}(\pi(\cdot \mid b_t) \parallel q_0(\cdot \mid b_t))]$ . A deterministic choice of action  $a$  at one decision has cost  $-\log q_0(a \mid b)$ . This does not make all state-only trajectory divergences equal to action divergences: actions can lead to indistinguishable states.

Kaufmann’s continuous-time result relates the corresponding path objective to stochastic model-predictive control under its diffusion assumptions. MTG actions are discrete; we use the belief-space formulation rather than claiming the Rust engine is a continuous diffusion.

**Information is valuable through subsequent decisions.** Information can be valuable when it lets the agent make better subsequent decisions. A planner that accounts for these contingent decisions can favour useful information-gathering actions without adding a separate information bonus. This interaction between acting and learning is called *dual control* (Fel’dbaum, 1960). Scrying and drawing cards remain worthwhile when their gameplay benefits outweigh their costs.



For example, casting a less important spell before committing a key threat both advances the position and elicits a response. If the opponent spends a counterspell, that response changes what the agent knows and how it should play next. The information matters because it enables a better follow-up, not simply because uncertainty decreases. In *closed-loop* planning, later actions depend on what is observed; a fixed, open-loop action sequence cannot express that contingency. A short or inaccurate planner can therefore miss useful information. This describes a capability to be evaluated, not a demonstrated property of our current planner.

**Why not the conventional factorisations.** The two EFE forms in common use are

$$G_{\text{IG}}(\pi) = -I_Q(\eta; o \mid \pi) - \mathbb{E}_Q[\log p_{\text{obs}}(o)], \quad (10)$$

$$G_{\text{AMB}}(\pi) = \text{KL}(Q(\eta \mid \pi) \parallel \tilde{p}_{\text{pref}}(\eta)) + \mathbb{E}_{Q(\eta \mid \pi)}[H[p_{\text{obs model}}(o \mid \eta)]]. \quad (11)$$

Here  $I_Q$  is expected mutual information (reduction in hidden-state uncertainty),  $H$  is Shannon entropy,  $p_{\text{obs}}$  is a preference over observations,  $\tilde{p}_{\text{pref}}$  a state preference, and  $p_{\text{obs model}}$  the observation likelihood. Observation preferences describe desired readings; likelihoods describe how readings arise. The factorisations have distinct assumptions and should not be treated as interchangeable objectives (Millidge et al., 2021; Kaufmann, 2026).

- **Information without instrumental value (“noisy TV”).** If uncertainty reduction earns an unconditional bonus, an agent can prefer learning facts that do not improve its later decisions. A stylised MTG test would offer repeated library inspection when the information cannot change any useful continuation, while using mana or forfeiting tempo needed to stabilise. Scry, surveil and cantrips are not inherently pathological: they can improve draws, supply cards or enable synergies. A failure is choosing them for the bonus *despite* their total effect on the game objective. Moreover, genuinely independent random noise has zero mutual information with the hidden state; a correctly calculated information-gain term need not reward it. Capture requires, for example, learnable but irrelevant variables or a misspecified novelty/uncertainty proxy. The test must distinguish these cases.
- **Avoidance of uncertain observations (“scotophobia”).** The term means an aversion to “dark”, poorly observed situations. The ambiguity term in (11) penalises entropy of observations *conditional on a state*, not simply the variance of game rewards or uncertainty about an opponent’s hand. Confusing these quantities can make informative or tactically favourable lines look costly merely because their feedback is uncertain. As a toy cost comparison, a line giving terminal costs  $-4$  with probability  $0.75$  and  $2$  with probability  $0.25$  has expected cost  $-2.5$ , better than a safe line costing  $-1$ . With equal action costs, an additional uncertainty surcharge above  $1.5$  reverses that ordering. Whether such a surcharge actually arises from the chosen observation model must be tested; unknown blockers or open mana alone do not establish it. Avoiding a risky line may instead be rational under loss-sensitive preferences.

These are **testable failure hypotheses, not observed MTG results**. The legacy ablation selectors use simplified entropy/likelihood proxies, not a validated exact implementation of every observation-channel factorisation. Section 6.3 must establish both the proxy’s meaning and the behavioural effect before attributing a game outcome to either mechanism.

**Soft selection rather than a hard minimum.** On a finite candidate set, let  $q_0(\pi)$  be reference probabilities and  $J(\pi)$  the predicted task cost of each candidate, before adding the KL regularisation for this selection step. Minimising  $\mathbb{E}_r[J] + \alpha^{-1} \text{KL}(r \parallel q_0)$  over selection distributions  $r$  yields

$$p_\alpha(\pi) = \frac{q_0(\pi)e^{-\alpha J(\pi)}}{\sum_j q_0(\pi_j)e^{-\alpha J(\pi_j)}}, \quad \text{softmax}_\alpha(J) = -\frac{1}{\alpha} \log \sum_i q_0(\pi_i)e^{-\alpha J(\pi_i)}. \quad (12)$$

This is again exponential reweighting. For equal priors, costs 0 and 1 and  $\alpha = 2$ , the lower-cost option receives probability approximately 0.88. As  $\alpha \rightarrow 0$  selection approaches the prior; as  $\alpha \rightarrow \infty$  it concentrates on minimum-cost candidates within the prior’s support. The scalar  $\alpha$  is inverse temperature (selection precision), distinct from JEPA’s  $\beta_J$ . It is not itself a measured compute budget: rollout count, horizon and wall-clock limits control computation separately. Nor should the same deviation cost be included in  $J$  and charged a second time through the KL regulariser.

**Where other signals belong.** Heuristic, graph and language-model scores can shape the reference action policy  $q_0$ : for example, a known combo completion may become a more probable default. The controller then evaluates whether predicted consequences justify following or departing from that default. Prior design still involves choices and calibration; replacing a weighted score sum with KL regularisation does not make these assumptions disappear. Below, the shorter symbol  $q$  denotes this action prior, never  $q_{\text{bel}}$  or the encoder distribution  $q_\phi$ . Current native KL pilots use analytic state projections and forecasts, not a trained JEPA transition model, and do not qualify the complete learning architecture.

### 3.4 Knowledge graphs as a symbolic prior

Cards form an inductive graph  $\mathcal{G} = (V, E, \tau)$  with typed edges  $\tau \in \{\text{combo}, \text{synergy}, \text{counter}, \text{enables}\}$  ingested from the OWL ontology and an embedding pipeline trained with GraphSAGE (Hamilton et al., 2017). GraphSAGE is inductive: a freshly printed card  $v_{\text{new}}$  obtains an embedding  $\mathbf{h}_{v_{\text{new}}} \in \mathbb{R}^{128}$  by sampling and aggregating its neighbours, without retraining. We exploit this in two ways: (i) the encoder feeds a graph latent  $g_t = \text{Enc}_{KG}(\mathcal{G}, s_t)$  into the recurrent dynamics so that the world model’s prior is graph-aware, and (ii) the passive prior  $q$  of (9) is tilted by graph distance to combo completions, mirroring GraphRAG’s “relational evidence” pattern (Edge et al., 2024).

In implementation, we separate immutable source facts (card oracle data, ontology-grounded edges) from learned extension evidence. Self-play enrichment appends provenance-linked learned nodes/edges instead of mutating base facts, allowing many agents and runs to contribute to one shared strategic memory while preserving reproducibility and auditability. Section 4.2 makes this separation a property of the schema rather than a convention.

### 3.5 The neuro-symbolic boundary

“Neuro-symbolic” is used loosely enough that it is worth stating exactly what is symbolic here, what is learned, and what crosses between them. The system contains three symbolic artefacts and three learned ones, and the interfaces between them are narrow by construction.

**Symbolic side.** (i) The **phase-rs** rules engine is a verified, deterministic transition system: it decides legality, resolves the stack, applies state-based actions, and is the only component permitted to mutate game state. (ii) The OWL 2 DL ontology (Section 4.2) fixes the vocabulary in which card semantics, turn structure, roles, and strategic constructs are expressed, and is checked by a reasoner and by SHACL shapes. (iii) The knowledge graph is the ABox populated against that TBox from Scryfall and combo databases.

**Learned side.** (i) A GraphSAGE encoder turns the graph into dense node embeddings. (ii) The JEPA world model learns latent dynamics from self-play trajectories. (iii) Language models supply a prior over the legal-action set.

**What crosses, and in which direction.** The crossings are the substance of the architecture:

1. **Symbolic  $\rightarrow$  neural (structure as inductive bias).** Graph structure is compiled into the encoder’s context channel  $g_t$ , and the legal-action set produced by the engine masks the controller’s output. The neural components therefore never have to learn legality or card identity; they learn only what is genuinely statistical.
2. **Symbolic  $\rightarrow$  neural (structure as reference measure).** The knowledge graph and the heuristic shape the passive prior  $q$  of (9). Symbolic knowledge thus enters the control objective as a *cost of deviation*, not as an additive score.
3. **Neural  $\rightarrow$  symbolic (induction of new relations).** Self-play traces are mined into candidate relations — card co-occurrence in winning positions, answer/threat pairs, archetype signatures — and written back as *induced* assertions. This is the direction most neuro-symbolic systems leave empty, and it is where the interesting failure modes live: a statistical regularity is not an ontological commitment.
4. **Validation as the gate between them.** Nothing induced is promoted to curated status automatically. Candidate relations are reified, weighted, attributed to the run that produced them, and validated by the shapes and reasoner before a human may accept them. This mirrors the finding of [Tiwari et al. \(2026\)](#) that the validator, not the generator, is what turns diverse neural proposals into a high-precision graph.

**Curated versus induced semantics.** The design commitment underlying all four crossings is that deterministic facts and learned regularities must remain distinguishable forever. A card’s oracle text is curated: it is definitional, immutable, and wrong is wrong. That two cards co-occur in 83 % of the games one of them wins is induced: it is defeasible, provenance-bearing, and may be an artefact of the deck pool sampled. Collapsing the two — writing a learned edge into the same relation as a curated one — destroys the ability to recover ground truth and makes the graph unauditable. Section 4.2 enforces the distinction with an `mtg:epistemicStatus` annotation on every axiom and a schema-level requirement that no induced assertion may exist without a `prov:Activity` that generated it.

### 3.6 General LLMs as indexed-action selectors

The action interface supplies a filtered observation  $o_t$  and an indexed engine-authored action list  $\mathcal{A}_t$ . A general chat LLM is asked to select an index:

$$\hat{i} = \text{LM}(\rho(o_t, \mathcal{A}_t)), \quad a_t = \mathcal{A}_t[\hat{i}]. \quad (13)$$

Only an index resolved to an original legal action can reach the engine. This restricts the action interface, not necessarily the model’s token generation: the current chat path parses a text reply rather than exposing a token-masked, calibrated probability distribution. Malformed replies and service failures need explicit handling and accounting. A low sampling temperature reduces output variability; it neither proves calibration nor represents a uniform-fallback probability.

### 3.7 Decision-tuned models as option selectors

A *decision model* here is a pretrained language model adapted to choose among supplied alternatives. It is not a world model: it returns a choice, not a predicted successor state or a simulated trajectory. Unlike a general chat selector prompted to emit an integer, its training and API are specialised for categorical decisions. Tev1 is an experimental Qwen3.5-derived family with supervised decision adaptation and a retained language-model output head ([Together AI, 2026](#); [Ollama, 2026](#)); decision tuning does not by itself establish MTG expertise.

Given a filtered observation  $o_t$ , a context projection  $R(o_t)$  and a subset  $\mathcal{S}_t \subseteq \mathcal{A}_t$  of engine-authored actions, it supplies

$$d_\psi(a \mid R(o_t), \mathcal{S}_t), \quad a \in \mathcal{S}_t, \quad (14)$$

where  $\psi$  denotes the selector parameters. The selected option maps back to the original action, preserving targets and object identity. Returned option probabilities indicate the model’s preference among the presented alternatives, not a probability of winning or a calibrated probability that a choice is correct.

The candidate generator is part of the policy: if a winning line is absent from  $\mathcal{S}_t$ , no reranker can recover it. Likewise, projection  $R$  can omit strategically important context. For example, selecting removal from a shortlist can be sensible only if the supplied state distinguishes the relevant threat, available mana and target. Measuring legal output is therefore insufficient; matched-context/candidate comparisons, shortlist coverage, tactical quality and inference cost are separate measurements (Section 6.2).

There are three possible uses, which should not be conflated: direct action selection, reranking candidates proposed by a planner, and supplying probabilities to the planner’s passive prior  $q_0$ . Only the first is currently integrated for Tev1. None replaces the rules engine or provides the latent transition model needed for JEPA rollouts. Local interface and terminal-game checks pass with a same-weight larger-context deployment; neither establishes calibrated choice quality or a Tev1 playing-strength advantage.

### 3.8 Self-play promotion as fixed-point iteration

Training combines all three signals. The replay buffer  $\mathcal{D}$  is collected by self-play between agent variants; the world model is fit on  $\mathcal{D}$ ; and the controller is improved iteratively by the champion-vs-challenger schema implemented in `scripts/train_pipeline.py`. At each iteration  $i$ , a challenger  $\pi_{i+1}$  is trained against  $\pi_i$  on the latest replay window and faces the champion in a best-of- $N$  match; promotion happens iff the empirical win rate exceeds 0.55. This is a stochastic-approximation fixed-point operator on the policy improvement step (Tesauro, 1995; Silver et al., 2017, 2018; Schrittwieser et al., 2020): under standard Robbins–Monro conditions it converges, in expectation, to a fixed point of the Bellman optimality operator restricted to the dynamics class  $\hat{T}$  that the world model has learned. Convergence in practice depends on  $\hat{T}$  being locally accurate, which we monitor by tracking JEPA validation loss across iterations.

## 4 System Architecture

### 4.1 Game engine

The runtime engine is `phase-rs` (Rust) ([phase-rs contributors, 2026](https://github.com/phase-rs/phase)), published at <https://github.com/phase-rs/phase> and accessed over its WebSocket interface via `src/integrations/phase_rs/`. Our Python side is intentionally adapter-only at execution time: `client.py` handles protocol envelopes, `runner.py` drives game loops, `adapter.py` maps legal actions/snapshots into our policy API, and `server_process.py` manages local server lifecycle for training scripts.

**Tracking a living upstream.** `phase-rs` is under active development and its wire protocol is versioned and enforced at handshake. Over the period covered by this report the upstream protocol advanced from v6 through v75 to the current v106 pin, which is a useful reminder that an external rules engine is a dependency with its own release cadence, not a fixed substrate. Two of those changes were structural rather than additive and required adapter work: rejections became a typed `ActionRejection` carrying a stable machine-readable code and a recovery disposition rather than a prose string, and the engine now publishes *viewer interactions* — an

| Comprehensive Rules concept           | CR section          | Implementing module                             |
|---------------------------------------|---------------------|-------------------------------------------------|
| Turn structure / phases / steps       | 500–514             | <code>src/engine/phases.py</code>               |
| APNAP priority & priority loop        | 116–117             | <code>src/engine/priority_loop.py</code>        |
| The stack & spell resolution          | 405–608             | <code>src/engine/stack.py</code>                |
| Layer system (continuous effects)     | 613                 | <code>src/engine/continuous_effects.py</code>   |
| Replacement effects                   | 614–616             | <code>src/engine/replacement_effects.py</code>  |
| Triggered abilities                   | 603                 | <code>src/engine/triggered_abilities.py</code>  |
| State-based actions                   | 704                 | <code>src/engine/state_based_actions.py</code>  |
| Combat (declare, block, damage steps) | 506–510             | <code>src/engine/combat.py</code>               |
| Mana & cost payment                   | 106, 601–602        | <code>src/engine/mana.py</code>                 |
| London mulligan                       | 103.5               | <code>src/engine/game_setup.py</code>           |
| Cleanup discard to hand size          | 514                 | <code>src/engine/phases.py::cleanup_step</code> |
| Loss from empty library on draw       | 104.3c, 704.5b      | <code>src/engine/state_based_actions.py</code>  |
| Legal-action enumeration              | — (engine-internal) | <code>src/engine/rules_engine.py</code>         |

Table 2: Mapping from Comprehensive-Rules concepts to the implementing engine module. Every cell is a real path; auditing a rule reduces to opening that file. The legal-action enumerator (`rules_engine.py`) is the only point of contact between the engine and any agent.

explicit, engine-authored protocol for the choices a player must make during resolution (targets, modes, damage assignment, ordering). The second is the more significant for us: it moves a large class of decisions out of the flat `legal_actions` list into a structured opportunity schema, which is strictly better for an agent that wants to reason about *what kind* of choice it is making. We pin the protocol constant in `client.py` and fail closed on mismatch rather than attempting to negotiate, on the principle that a silently misinterpreted game state is worse than a refused connection.

Core integration modules:

- `client.py`: protocol handshake, message parsing, and typed message dataclasses for `GameStarted`, `StateUpdate`, and `GameOver`;
- `runner.py`: end-to-end game loop for a Python-controlled seat, including structured per-decision JSONL traces;
- `adapter.py`: practical `GameAction` ↔ `Action` bridge and snapshot projection for native Python policies;
- `agent_bridge.py`: random/heuristic/LLM pickers plus adapter-backed native-agent picker support;
- `server_process.py`: managed local `phase-server` process startup/shutdown for reproducible training/evaluation runs.

The old Python engine remains in-repository as legacy compatibility code, but new experiments and reported benchmarks use `phase-rs` as the authoritative runtime transition system.

Table 2 maps each Comprehensive-Rules concept to the exact module that implements it; readers reproducing the system can use it as a reading order through the codebase.

## 4.2 Symbolic substrate: the ontology

The symbolic layer is an OWL ontology of MTG derived from the Comprehensive Rules and aligned to the DOLCE-UltraLite foundational ontology, with OWL 2 DL as its modelling target. It is a contribution in its own right and is described in a companion paper (Neubürger and

Neubürger, 2026); we summarise here the intended symbolic contract and its current qualification boundaries.

Three properties matter for the argument of this paper. First, the ontology is *expressive enough to be wrong*: logical analysis exposed incompatible property characteristics that were repaired, and Hermit finds the import-resolved schema consistent with no unsatisfiable named classes. This is not complete OWL 2 DL profile conformance of the raw external documents, nor certification of a populated card graph. The CR 205.3 card-type/subtype correlation is also checked procedurally. Second, its factual vocabulary is *derived, not curated by hand*: 3,165 source rules are parsed and 830 vocabulary subjects carry direct document provenance; rule text remains local rather than being redistributed. Third, and most importantly here, it distinguishes *curated* from *induced* assertions at the schema level. That distinction supplies a necessary audit boundary for the learning loop of Section 4.3.

For orientation: cards are imported from Scryfall as `CardDesign` nodes with oracle text and an SBERT embedding; mechanics, abilities, archetypes and combos are first-class nodes; APOC supplies PageRank, community detection and shortest-path queries; and multi-hop retrieval over this structure is the “relational evidence” pattern of GraphRAG (Edge et al., 2024), which suits questions — *which two cards in my library combo, and does either die to a counterspell?* — that similarity search answers badly.

### 4.3 Playing to build the graph

The knowledge graph is usually treated as a fixed input: someone curates it, the agent reads it. We are interested in the other direction as well, because an agent that plays thousands of games observes something a curator cannot — which cards actually co-occur in positions that get won, against which opponents, at which points on the curve.

The loop has four steps. Agents play on the rules engine and emit structured decision traces. Traces are post-processed into candidate relations: card pairs recurring in winning positions, answer/threat pairs, archetype signatures. Candidates are written into the graph as *induced* assertions — reified, weighted by support and refutation counts, and attributed under PROV-O to the run that produced them. Later agents query those assertions as an additional prior, and their play produces the next round of evidence.

Two design commitments keep this from degenerating.

**The extension layer is append-only and typed apart.** An induced synergy is not an edge in the curated `synergisesWith` relation; it is a reified evidence node that the schema requires to be attributable to an activity. A consumer needing deterministic ground truth recovers it by filtering on epistemic status. This matters because the two kinds of claim have different logic: a card’s oracle text is definitional and wrong is wrong, whereas a co-occurrence statistic is defeasible and may be an artefact of the deck pool sampled. Systems that write learned edges into curated relations lose the ability to tell the difference, and with it the ability to audit.

**Promotion is not automatic.** Nothing is reclassified from induced to curated without a human. We regard the promotion gate — what support threshold, what handling of disagreement between runs (Ehrenmüller et al., 2026) — as an open problem rather than a solved one, and say so in Section 7.

**Agent reasoning is itself in the graph.** A decision is recorded as a `prov:Activity` carrying the objective decomposition it was selected by, the belief it was taken under, and the policy precision in force. “Why did seat 0 attack on turn 6?” is therefore a graph query rather than a log grep. This is a deliberate response to the observation (Sabou, 2026) that trustworthy neuro-symbolic systems need explanations grounded in domain knowledge with provenance



attached, rather than post-hoc attribution scores; it also means the explanation and the game state it refers to are expressed in one vocabulary.

**Collective intelligence, stated carefully.** The attractive reading is that heterogeneous agents accumulate shared strategic memory across runs, so that later agents inherit what earlier ones discovered without retraining. The framework supports that reading structurally. We have not yet demonstrated it: showing that graph enrichment from run  $n$  measurably improves agents in run  $n+1$ , and is not merely overfitting to a fixed deck pool, is the experiment that would substantiate the claim, and it is not among the results reported here.

#### 4.4 World model

Let  $o_t$  be the perspective-filtered observation at decision step  $t$  and  $g$  the player’s KG context (deck, archetype tag, known opponent information). The world model decomposes as:

$$z_t = E_\phi(o_t, g_t) \quad (\text{observation encoder}) \quad (15)$$

$$g_t = \text{Enc}_{\text{KG}}(g, o_t) \quad (\text{symbolic context encoder}) \quad (16)$$

$$z_{t+1}^M, h_{t+1} \sim M(z_t, a_t, h_t) \quad (\text{recurrent stochastic dynamics}) \quad (17)$$

$$\hat{z}_{t+1} = f_\theta(z_t, a_t) \quad (\text{JEPA predictor head}) \quad (18)$$

$$a_t = C(z_t, g_t, h_t) \quad (\text{controller}). \quad (19)$$

$E_\phi$  produces a 256-dimensional representation of the visible position and optional symbolic context. The JEPA objective is (3): next-embedding prediction plus encoder regularisation against a standard normal distribution. The recurrent mixture-density dynamics, reward head and controller are separate from that predictor; their use in dream search does not imply an already-qualified JEPA-driven planner (Section 3.2).

The prediction residual  $\|\hat{z}_{t+1} - z_{t+1}\|_2$  measures disagreement in the chosen representation. It may flag unfamiliar transitions for analysis, but does not distinguish model error, hidden-state uncertainty and useful tactical novelty by itself. In particular, it is not an intrinsic reward in the belief-space KL objective. The encoder also supports a separate feature-reconstruction training path; this does not make reconstruction of a complete successor state part of the JEPA loss.

#### 4.5 Combining selectors with a planner

The intended integration separates *what is a plausible default action* from *what its predicted consequences are*. Heuristic, graph or model preferences can supply  $q_0(a \mid b_t)$ ; a transition model and an outcome cost supply the planning component of (9). The earlier weighted-fusion policy is retained as a separate baseline, not identified with the KL controller. Language-model scores must be calibrated and traced if used as probabilities, and a deck’s preferred curve is a strategic heuristic, not a hard legality constraint.

**Illustrative trade-off.** Suppose a prior gives probabilities 0.6 to countering a spell and 0.4 to passing. Their deterministic one-step deviation costs are  $-\log 0.6 \approx 0.51$  and  $-\log 0.4 \approx 0.92$ . If predicted outcome costs are 0.8 and 0, the totals are approximately 1.31 and 0.92: the predicted advantage of passing overrides the prior’s preference for countering. These invented numbers illustrate the decomposition, not model outputs from a recorded game. Obtaining reliable outcome forecasts and context-conditioned priors is the experiment, rather than a consequence of writing down the objective.

## 4.6 Self-play training loop

We use a champion-vs-challenger promotion scheme:

---

**Algorithm 1** Champion-vs-Challenger Self-Play (Phase A.5).

---

```

1: Initialise pool  $\mathcal{P}$  with  $K$  randomly-seeded agents.
2: Pick champion  $c^* \leftarrow \arg \max_{c \in \mathcal{P}} \text{ELO}(c)$ .
3: for round = 1, ...,  $R$  do
4:   Sample challenger  $c \sim \mathcal{P} \setminus \{c^*\}$ .
5:   Play  $N$  games  $c$  vs  $c^*$ , collect trajectories  $\tau$ .
6:   Update ELO for both; train all pool members on  $\tau$ .
7:    $w_r \leftarrow$  challenger win rate over the  $N$  games.
8:   if  $w_r \geq \theta$  then
9:     Promote:  $c^* \leftarrow c$ ; bump ELO by +50, demote former by -20.
10:  end if
11: end for

```

---

$\theta$  is the promotion threshold (default 0.55). The loop is implemented in `src/training/r1_trainer.py`; `scripts/train_pipeline.py` is the execution entry point, not evidence of a qualified end-to-end native promotion experiment.

## 4.7 Agent zoo: algorithmic specification

The framework combines Python agents satisfying `decide_action(state, legal_actions) → action` with native `phase-rs` pickers selecting an index into the engine’s legal-action list. They span the spectrum from combinatorial baselines to the V+M+C+JEPA+KG+LLM fusion of Section 4.4; the native pickers additionally expose KL control and decision-tuned language models. We summarise their algorithmic roles below; implementation detail belongs in the companion technical report.

**Random policies** ( $\pi^{\text{rand}}$ ). The native-wire baseline used in the experiments samples uniformly from the supplied legal-action list using an instance-local seeded RNG. It does not evaluate cards, targets or future consequences; if concession is offered, it is part of that sampling domain. The separate translated Python random agent is a different policy: a weighted sampler with  $w_{\text{land}} = 25$ ,  $w_{\text{cast}} = 10$  (30 for command-zone casts),  $w_{\text{atk}} = 8$  and  $w_{\text{pass}} = 1$ , using a process-global RNG. Results for these two implementations must not be pooled under one label.

**Heuristic** ( $\pi^{\text{heur}}$ ). Deterministic priority cascade

$$a^* = \arg \max_{a \in \mathcal{A}_{\text{legal}}} \text{prio}(a), \quad \text{prio} : \begin{cases} \text{PLAY\_LAND} \mapsto 100 \\ \text{CAST\_SPELL} \mapsto 90 \\ \text{DECLARE\_ATTACKERS} \mapsto 80 \\ \text{DECLARE\_BLOCKERS} \mapsto 70 \\ \text{ACTIVATE\_ABILITY} \mapsto 60 \\ \text{PASS} \mapsto 1 \\ \text{CONCEDE} \mapsto 0, \end{cases}$$

with three tactical refinements: (i) counter-spell discipline filters `counter target` casts unless an opponent spell tops the stack, (ii) when `prefer_aggressive` is set the agent picks an attack at the top tier, (iii) on the blocking step it chump-blocks the largest unblocked attacker with the smallest blocker. Tie-breaking uses an instance-local seeded RNG. This describes the translated

Python heuristic, not the native-wire baseline: the latter prefers land, cast, activate, then attack, breaks within-tier ties randomly, and otherwise passes or chooses a non-concession action. It does not rank targets or simulate combat.

**Built-in opponent: phase-ai.** The upstream engine includes a distinct decision system (phase-rs contributors, 2026), not a renamed version of either Python baseline. It constructs legal candidates, applies tactical policies and safety/loop guards, and scores positions using life, board material, hand resources, zone quality and synergy. Its turn-dependent evaluation weights are distributed as fitted parameters; our experiments neither retrain nor independently validate the upstream fitting procedure. Difficulty controls stochastic selection, tactical lookahead and search. The **VeryEasy** preset used here disables tree search but retains scored tactical decisions and softmax selection with temperature 4; it is *not uniform random*. For comparison, the native **Medium** preset enables beam-plus-rollout search with depth 2, a 24-node ceiling, branching cap 5 and one depth-1 rollout sample, with a 1.5-second interactive search deadline. Special decisions may take deterministic policy paths rather than the generic softmax path.

An important asymmetry concerns information, not just compute. The native AI receives the internal game state; the shipped search presets set hidden-zone determinization samples to zero and use the unsampled state. Our external policies receive perspective-filtered observations. Consequently native-AI performance is an *anchor under its declared information regime*, not a fair partial-information algorithm ranking. Matched-information comparisons require masking and independently checking the native opponent’s access. The opponent is held fixed during a learning experiment and does not update from our gameplay traces.

| Policy             | Decision rule                                | Card/tactical reasoning                        | Search and learning                          |
|--------------------|----------------------------------------------|------------------------------------------------|----------------------------------------------|
| Wire random        | Uniform over offered legal actions           | None                                           | No search or updates                         |
| Wire heuristic     | Fixed action-type tiers; seeded tie sampling | No target or combat valuation                  | No search or updates                         |
| Python heuristic   | Priorities plus counter/attack/block rules   | Limited hand-crafted tactics                   | No search or updates                         |
| phase-ai, VeryEasy | Tactical scoring; stochastic choice          | State, card, combat and synergy policies       | No tree search; fixed fitted evaluator       |
| phase-ai, Medium   | Tactical scoring plus beam/rollout           | Same native policies and internal-state access | Bounded native search; no study-time updates |

Table 3: Baseline implementations are not interchangeable. Seeded external selection does not seed engine shuffling or native-AI selection. Native internal-state access differs from external perspective filtering.

**KG-Heuristic** ( $\pi^{\text{KG+heur}}$ ). Inherits  $\pi^{\text{heur}}$  but, when more than one `CAST_SPELL` is legal, re-ranks them through the knowledge graph. Let  $H$  be the agent’s hand and  $B$  its battlefield. For each candidate cast  $c$ ,

$$\text{score}_{\text{KG}}(c) = 10 \cdot \mathbf{1}[c \in \text{NearComboMissingPieces}(H \cup B)] + \sum_{p \in B} \text{strength}(\text{KG-edge}(c, p)),$$

where the indicator is computed by a single Cypher round-trip (`kg.detect_near_combos`) and synergy strength comes from the `HAS_SYNERGY` edges of the knowledge graph. The best-scoring cast replaces the cast subset in the legal-action list and the parent priority logic resumes, so `PLAY_LAND` can still beat `CAST_SPELL`. On any KG exception the agent latches `kg_disabled = True` and silently degrades to  $\pi^{\text{heur}}$ .

**Ollama LLM** ( $\pi^{\text{LLM}}$ ). The LLM is consulted as a discrete posterior over the legal-action set, never as a free-form generator. Concretely, for each decision we serialise the board state  $s_t$  (life totals, hand sizes, lands, creatures) and an enumerated list  $\{(i, a_i)\}_{i=0}^{|\mathcal{A}_{\text{legal}}|-1}$ , prompt the model to “respond with ONLY the option number”, parse an integer  $\hat{i}$  from the reply, and return  $a_{\hat{i}}$ . Formally,

$$\hat{i} = \pi_{\text{LLM}}(\text{prompt}_t), \quad a^* = a_{\hat{i}} \quad \text{if } 0 \leq \hat{i} < |\mathcal{A}_{\text{legal}}|.$$

Only an in-range index reaches the engine; invalid replies are not clipped to a legal index. Native chat selection uses temperature 0.2, disables thinking and caps output at 16 tokens, with a 30-second request timeout. Single-action choices bypass inference. HTTP/connection failures, invalid replies and substituted concessions are explicitly recorded as fallback events in decision traces. A completed game therefore need not consist solely of language-model decisions.

**Decision-model deployment** ( $\pi^{\text{dec}}$ ). The conceptual distinction from chat selection and latent dynamics is given in Section 3.7. The current deployment uses Tev1 0.8 B/4 B through Ollama’s structured option-selection interface.

For a filtered observation  $x_t$ , let  $\tilde{x}_t = R(x_t)$  be its decision context projection,  $\mathcal{A}_t$  the engine-authored legal-action set and  $\mathcal{S}_t = C_k(x_t, \mathcal{A}_t)$  a candidate set,  $|\mathcal{S}_t| \leq k \leq 24$ . The selector supplies a categorical preference

$$p_{\theta}(a \mid \tilde{x}_t, \mathcal{S}_t), \quad a \in \mathcal{S}_t,$$

and its selected key is mapped back to the *original* engine action, including targets and object identities. The first implementation uses heuristic action tiers with seeded tie-breaking, excludes concessions when alternatives exist, and retains a pass option if present. A forced choice bypasses inference. The 24-option cap follows the model’s reported training range, rather than implying that MTG’s action space has that size. We log the candidate indices, model probabilities and  $|\mathcal{S}_t|/|\mathcal{A}_t^{\text{eligible}}|$ . This fraction measures *action coverage*, not recall of the best move. Inference failures are explicit incomplete runs; no random-policy fallback is credited to the decision model.

Input budgeting is also an interface constraint. Local Ollama 0.35.1 rejects request bodies above 64 KiB. The installed Tev1 tag initially sets its context parameter to 2,050 tokens; request-level overrides do not change that setting in the structured endpoint. An isolated derived tag retains the same model weights and sets the context to 8,192 tokens. A 2,906-token structured request succeeds with this tag without truncation. This is a deployment-configuration change, not additional model training; extended-context decision quality must be evaluated. Raw engine snapshots exceed the body limit. Our versioned card-aware projection retains visible identities and instance IDs, public characteristics, life/mana and decision state, grouping equivalent card rows without merging instance identities. It omits hidden placeholders, library order, internal journals and compiled rule syntax trees; it is not a lossless rules-state encoding. Projection version and request bytes are recorded, and oversize inputs fail explicitly rather than being silently cropped. The configured context remains finite; larger positions can still fail explicitly. Context configurations and failure accounting must be held fixed within a comparison.

This component can replace a chat selector at the action boundary without replacing the rules engine, the world model or KL planning. Using it as a critic over world-model candidates, or as a learned passive prior, is a separate proposed integration. Our hypothesis is improved selection reliability and a favourable strength–latency trade-off, not that decision tuning already makes Tev1 a stronger MTG player.

**Decision 2.0: a wider categorical action domain.** The Decision 2.0 family ([vLLM Semantic Router Team, 2026](#)) provides a distinct structured classifier interface, not a chat completion model. Its released choice contract accepts 2–255 options; Eos 0.8B specifies a 16,384-token input budget, while Kai 0.6B specifies 8,192 tokens. These limits widen the offered

decision domain but do not remove MTG’s combinatorial action space: larger legal sets still require a declared candidate-selection rule. The adapter reuses perspective filtering, card-aware projection, original action identities and probability validation, while allowing the wider option cap. Over-budget inputs and model errors remain explicit failures.

The reference interface is local Transformers `system_one`, with a pinned weight and custom-code revision; a vLLM inference server is not required merely because the publishing organisation is named vLLM Semantic Router. The classifier head must be retained: converting the backbone alone to a chat model does not reproduce this decision interface. Choice-set ablations at 24, 64, 128 and 255 options should report coverage, context budget, latency and order sensitivity separately. Generation-to-generation comparisons must also account for precision and hardware: the existing Tev1 deployment uses Q8 weights, whereas this reference CPU runtime uses FP32. Publisher GPU latencies are not measurements of our CPU deployment, and released classification scores are not MTG playing-strength results.

Real Eos loading currently fails on the Windows evaluation host before any choice is made: the upstream checkpoint fingerprint serializes relative paths with platform-specific separators, whereas the published identity uses slash-separated keys. Every actual model/tokenizer file hash matches the manifest, and canonical slash-key serialization reproduces its identity. The integrity check is not disabled. A separate local Linux CPU check loads the unchanged package with Torch 2.11.0 and Transformers 5.17.0, four threads and a 6 GiB memory limit. Cold preparation takes 53.7 s; 30- and 255-option requests return original option keys and complete finite normalized probability maps, using 1,136 and 9,771 input tokens respectively, without generated text or truncation. Single-request forward times are 12.2 and 136.1 s, not a latency benchmark or publisher GPU timing. The synthetic fixture asks for the largest JSON-encoded amount, but neither output selects that maximum. Interface validity is therefore not decision quality. No Decision 2.0 MTG episode, larger variant or GPU result is claimed; direct Windows deployment still needs a portable runtime revision.

**Active Inference / KL control ( $\pi^{\text{KL}}$ ).** Maintains a categorical belief  $b_t$  over hidden-information hypotheses about the opponent’s hand (`holds_removal`, `holds_counterspell`, `holds_threat`, `holds_nothing`). Control costs are charged against a passive prior, while analytic forecasts supply outcome costs. Two planners are available: a belief-tree reference planner and a sampled MPPI-style approximation. Current sampled continuations use averaged stage potentials and soft aggregation, not an exact calculation of the full trajectory objective. Defaults are  $H = 3$ ,  $K = 48$  and selection precision  $\alpha = 4$  (called `beta` in configuration). Belief entropy is reported for diagnostics and is not an additive reward; this excludes that particular bonus mechanism, not every possible model-induced failure.

**Preference potential.** Goals use  $V(\eta) = -\log p_{\text{pref}}(\eta)$  from (6), with costs for life/material differences, turn pressure and terminal outcomes. These coordinates make the goal specification inspectable, but its hand-set weights and incomplete state projection may still misvalue a position. Anchoring goals to game consequences avoids explicitly rewarding sensor readings; it is not a universal correctness guarantee.

**World-Model agent ( $\pi^{\text{WM}}$ ).** Supports the V+M+C+JEPA design of Section 4.4. With a qualified graph encoder, the intended observation representation is

$$z_t = E_\phi(o_t, g_t), \quad g_t = \text{Enc}_{\text{KG}}(g, o_t), \quad h_t \text{ retains recurrent history.}$$

Two action-selection modes are available: a fast direct policy over the supplied legal actions, and dream search that compares predicted cumulative rewards from recurrent latent continuations, as in (4). The native feasibility configuration supplies card embeddings and an explicit checkpoint, but not live relational retrieval. The JEPA head is trained separately; selecting a JEPA-trained

checkpoint does not establish that this search unrolls the JEPA predictor or that its recurrent dynamics are trained. Imagined continuations are approximate and do not receive fresh engine legality checks.

**LLM-Fusion agent** ( $\pi^{\text{fuse}}$ ). A weighted-fusion baseline combining four scalar signals per legal action,

$$\text{score}(a) = \alpha_h s_{\text{heur}}(a) + \alpha_g s_{\text{KG}}(a) + \alpha_w s_{\text{WM}}(a) + \alpha_\ell \log p_{\text{LLM}}(a), \quad a^* = \arg \max_a \text{score}(a),$$

with default weights  $(\alpha_h, \alpha_g, \alpha_w, \alpha_\ell) = (0.15, 0.15, 0.40, 0.30)$ . The LLM signal is skipped when one action dominates the heuristic-plus-WM signal by more than a threshold of 0.8. The skipped-call rate must be measured, not assumed. Setting a component’s weight to zero defines a removal intervention only if the full condition actually activates that component; the native fusion feasibility run does not supply graph retrieval.

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**Algorithm 2** LLM-Fusion decision (one priority window).

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**Require:** state  $s_t$ , legal actions  $\mathcal{A}$ , world model  $(\text{Enc}_V, M, C)$ , KG  $\mathcal{G}$ , opponent model  $b_t$ , weights  $(\alpha_h, \alpha_g, \alpha_w, \alpha_\ell)$

- 1:  $\mathbf{s}_h \leftarrow \text{HeuristicScores}(s_t, \mathcal{A})$
- 2:  $\mathbf{s}_g \leftarrow \text{KGScores}(s_t, \mathcal{A})$  ▷ combo + synergy bonuses
- 3:  $z_t \leftarrow \text{Enc}_V(s_t)$ ;  $\mathbf{s}_w \leftarrow \text{DreamSearch}(z_t, h_t, \mathcal{A}; K, D)$
- 4: **if**  $\text{spread}(\mathbf{s}_h + \mathbf{s}_w) > \tau_{\text{obvious}}$  **then**
- 5:      $\mathbf{s}_\ell \leftarrow \mathbf{0}$  ▷ skip the LLM call
- 6: **else**
- 7:      $\mathbf{s}_\ell \leftarrow \log p_{\text{LLM}}(\cdot \mid \text{prompt}(s_t, \mathcal{A}))$
- 8: **end if**
- 9: **return**  $a^* \leftarrow \mathcal{A}[\arg \max_i (\alpha_h \mathbf{s}_{h,i} + \alpha_g \mathbf{s}_{g,i} + \alpha_w \mathbf{s}_{w,i} + \alpha_\ell \mathbf{s}_{\ell,i})]$

---

**Reasoning traces.** Every agent writes a `ReasoningTrace` dict to `self.last_reasoning` before returning. The orchestrator flushes these to `runs/<out>/logs/game_MNNN.jsonl`, one line per decision, schema {turn, phase, player\_id, agent\_kind, rationale, legal\_action\_count, chosen\_index, top\_candidates, scores, beliefs}. This is the canonical telemetry format consumed by the post-hoc analysis notebooks and by the Streamlit visualisation panel discussed in Section 8.

## 5 Implementation

The scientific object of interest is the *interaction* between symbolic structure and learned planning, not the package layout. The artifact supplies a native rules runtime, an ontology and card-data pipeline, trainable latent-model components, and selectors combining heuristic, model and language signals. These components do not yet constitute a qualified native training-and-retrieval loop. The proposed interventions isolate their contributions after activation and calibration have been verified: MTG provides a setting for studying how symbolic priors, learned dynamics and uncertainty-aware action selection cooperate.

**Reproducibility.** The current `phase-rs` benchmark records the repository and engine checkout revisions, the actual server identity from the handshake, protocol version, deck fingerprint, full configuration and per-game picker seed. Candidate selection uses a seeded Python RNG. Engine shuffles, opponent-AI randomness and model-service inference are not all controlled by that seed; matched picker seeds are therefore not deterministic paired-game replays. Decklists are committed under `data/decks/`. Engine RNG control and seat rotation remain required for a fully paired comparison.



## 6 Evaluation Design and Current Evidence

We distinguish measured native execution from the controlled experiments needed to establish learning and graph-mediated improvement. All empirical results in this section refer to the current native runtime. The research questions and ablations specify falsifiable comparisons; they are not presented as completed learning experiments.

### 6.1 Research questions and staged ablation design

The central empirical claim to investigate is not merely that heterogeneous agents can execute legal actions. It is that agents can improve through experience, that different decision policies offer measurable effectiveness–cost trade-offs, and that strategic evidence mined from play can help later agents. We therefore separate five questions:

1. **RQ1 (policy comparison):** how do qualified playing policies compare in outcomes, completion and inference cost under declared deck/opponent conditions?
2. **RQ2 (parametric learning):** does training on native gameplay improve held-out predictions and downstream play compared with the same architecture before training?
3. **RQ3 (graph memory):** does training-induced strategic evidence improve a frozen consumer beyond curated relations alone?
4. **RQ4 (complementarity):** do weight learning and graph learning contribute independently, or is there an interaction?
5. **RQ5 (transfer):** can evidence from one producer policy improve another policy on held-out decks, and does cumulative memory generalise?

These are planned questions, not five established contributions. A null or harmful graph effect is a legitimate result; failure to activate the graph is an integration failure, not a null effect.

| Question         | Ablation conditions                                             | Held fixed                                                 | Principal evidence                                      |
|------------------|-----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------|
| RQ1: policy      | Random, heuristic, chat, decision-tuned, model-based, control   | Declared decks, opponent, information and inference budget | Outcome/completion effects and strength–cost frontier   |
| RQ2: weights     | Initial; predictive; policy; joint training                     | Architecture, graph, experience and optimisation budget    | Held-out prediction and play across training seeds      |
| RQ3: memory      | None; curated; induced; shuffled graph                          | Consumer weights, card facts, retrieval/context budget     | Induced-minus-curated effect and retrieval activation   |
| RQ4: interaction | $2 \times 3$ weights/graph factorial; focused component removal | Splits, opponent, candidate and compute budgets            | Weight/graph difference-in-differences with uncertainty |
| RQ5: transfer    | Own; other; mixed producer; accumulated snapshots               | Frozen consumer, experience exposure, extraction rules     | Held-out deck-family and cross-producer effects         |

Table 4: Research-question-driven ablation overview. A policy-stack comparison is not an isolated mechanism ablation. Weight-learning and graph-learning conditions require qualified native training and retrieval; they are experimental specifications, not completed results.

**Study A: policies before mechanisms.** First qualify random/heuristic, general chat, decision-tuned, explicit-checkpoint model-based and control stacks against an unchanged native-AI anchor. Deployment comparisons disclose differing contexts and candidate generators; mechanism comparisons hold those inputs fixed. A claim about decision tuning requires a size-matched base/chat comparator, identical options and option-order controls. A KL-versus-EFE objective claim requires matched models, candidate sets, horizons and compute. Native-AI episodes are not an agent-versus-agent tournament: the latter needs a separately qualified multi-connection controller, independently filtered observations and seat/start-player balancing. Matchup tables and strength–cost frontiers are more informative than assuming a transitive Elo ranking.

**Tournament reporting and Buchholz.** In a future Swiss tournament, agents play only a subset of opponents, so equal match points can reflect different schedule difficulty. With three points for a win, one for a genuine draw and zero for a loss, let  $P_j$  be opponent  $j$ ’s final match points. The Buchholz tiebreak for agent  $i$  is

$$B_i = \sum_{r \in \mathcal{R}_i} P_{\text{opp}(i,r)}, \quad (20)$$

where  $\mathcal{R}_i$  contains its played rounds: an opponent encountered twice contributes twice. For example, opponents finishing with 6, 9 and 12 points yield  $B_i = 27$ . Among agents tied on their own points, a larger Buchholz value indicates a higher-scoring opponent schedule, not a larger win probability or an estimate of causal policy strength. Byes and forfeits need a declared convention; operational failures are not silently scored as genuine draws. Our proposed convention excludes unplayed rounds from this sum and reports them separately. Balanced head-to-head outcome tables and uncertainty remain the primary comparison; no current native Swiss standings are claimed.

**Study B: demonstrate learning rather than checkpoint loading.** Separate native games into training, validation and final test partitions before any extraction or optimisation. Compare initial weights, predictive training, policy training and a qualified joint-training condition using declared data/optimisation budgets. Replicate initialisation/training runs and evaluate prespecified training milestones; select checkpoints on validation, not the final holdout. Report both downstream game outcomes and prediction diagnostics: reward/terminal calibration, action sensitivity, latent collapse and multi-step error. Embedding errors from independently changing latent spaces are not directly comparable without a common evaluation representation. The current one-step JEPA objective and recurrent dream-search path are distinct; a JEPA-planning ablation needs a qualified connection between latent prediction, reward learning and action evaluation.

**Study C: an intervention on graph memory.** Freeze consumer weights and immutable card facts. Compare no relational retrieval ( $K_{\text{none}}$ ), curated relations ( $K_{\text{cur}}$ ), curated plus training-induced evidence ( $K_{\text{ind}}$ ), and curated plus type/degree/support-matched shuffled evidence ( $K_{\text{shuffle}}$ ). Hold non-relational card information, retrieval/token budgets and candidate actions fixed. The principal contrast is  $K_{\text{ind}} - K_{\text{cur}}$ ; the shuffled arm controls extra graph volume and context. More triples or winning-deck co-occurrence alone do not demonstrate useful strategic knowledge.

Induced evidence retains generating run/game/deck, extraction version, support/refutation and opportunity counts, with deck context to expose co-occurrence confounding. Retrieval traces identify which evidence was used and how it affected scoring. Blinded relation review and removal probes support mechanism interpretation, but observed associations remain defeasible. No test-game evidence is inserted before the locked evaluation finishes.

**Study D: a minimal factorial for complementarity.** Instead of every possible model combination, cross initial versus trained weights ( $W_0, W_1$ ) with the three principal graph conditions: For a prespecified game-level estimand  $Y$ , the interaction is

|                 | $K_{\text{none}}$     | $K_{\text{cur}}$     | $K_{\text{ind}}$     |
|-----------------|-----------------------|----------------------|----------------------|
| Initial weights | $W_0/K_{\text{none}}$ | $W_0/K_{\text{cur}}$ | $W_0/K_{\text{ind}}$ |
| Trained weights | $W_1/K_{\text{none}}$ | $W_1/K_{\text{cur}}$ | $W_1/K_{\text{ind}}$ |

Table 5: Planned  $2 \times 3$  learning/graph factorial. Eligibility requires real training and graph interventions, not optional modules left inactive.

$$\Delta_{\text{int}} = [Y(W_1, K_{\text{ind}}) - Y(W_1, K_{\text{cur}})] - [Y(W_0, K_{\text{ind}}) - Y(W_0, K_{\text{cur}})]. \quad (21)$$

Report its uncertainty rather than declaring synergy from a high full-stack point estimate. Subsequently remove LLM reranking, graph retrieval, trained forecasts and planning one at a time in a qualified fusion consumer. Fixed-weight removal and validation-retuned removal answer different questions; equal-compute and natural deployment-cost comparisons are also separate.

**Study E: transfer and cumulative evidence.** Generate graphs from different producer policies under equal experience exposure and extraction rules. A frozen consumer receives its own-policy, another producer’s, mixed-producer or curated-only graph. Evaluate held-out deck families with explicit card/deck overlap audits; multiple burn lists are not independent strategic archetypes. Freeze successive graph snapshots after training-only batches, use validation to monitor accumulation, and reserve an untouched final test for the locked final snapshot. Separate graph-only accumulation from joint weight/graph updates and replicate independent producer runs: successive snapshots from one run are correlated.

**Shared validity and publication boundaries.** Whole games, deck families and producer batches define splits; decision-row splitting leaks both trajectories and induced relations. Deployed retrieval must respect perspective filtering. Before strength runs, require at least 20 calibration games per intended condition, at least 95 % completion, zero protocol/illegal-choice failures and audited action semantics. Prespecify principal contrasts, samples, exclusions and stopping rules. Report wins/completed, completion and wins/scheduled together, retaining genuine draws separately from operational failures. Cluster uncertainty over training/graph-producer runs as well as evaluation games and stratify by deck/opponent; correct the confirmatory contrast family for multiplicity. A roughly five-percentage-point two-arm effect near 50 % needs about 1,570 independent games per arm for 80 % power at two-sided 5 %, before clustering or multiplicity; the factorial requires simulation-based power. These are planning requirements, not authorisation for an unlimited campaign.

Component-activation and loaded-parameter audits are mandatory: a missing checkpoint, random initialisation, heuristic fallback or unused graph cannot be credited to the named full model. The core study must substantiate policy comparison, learning and graph memory; cross-agent accumulation and Commander are extensions only after their independent gates pass.

## 6.2 Decision-model benchmark protocol

**What is being tested.** The unit of deployment is one controlled seat against a fixed **phase-ai** difficulty, deck and rules-engine revision. We compare heuristic selection, general chat-model selection, Tev1 0.8 B/4 B and belief-space KL control. The question is whether decision adaptation

improves legal-action selection reliability and playing strength per unit of inference cost. Legal output alone is not evidence of tactical quality. The available episode harness is a built-in-AI opponent benchmark, not yet an agent-vs-agent round robin or a four-player Commander evaluation.

| Comparison          | Scientific question                                            | Required controls                                                                                                        |
|---------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Deployment stacks   | Which complete policy offers the best strength–cost trade-off? | Fixed engine, decks, difficulty, hardware; disclose candidate and context differences.                                   |
| Matched rerankers   | Does decision tuning improve choice quality?                   | Same saved observations, candidate identities, descriptions and option order; size-matched Qwen3.5 chat/base comparator. |
| Shortlist ablation  | How much does the candidate generator determine the result?    | Vary $k$ and candidate source; report coverage, action-type omissions and order sensitivity.                             |
| Planner integration | Does the selector help world-model candidates or KL priors?    | Hold rollout model, horizon and compute budget fixed; change only the selector/prior.                                    |

Table 6: Planned decision-model comparisons. Only the deployment adapter, tracing and episode harness are implemented; the controlled reranker and planner-integration experiments are follow-up work.

**Measurements and failure accounting.** The implemented harness reports completed-game win rate with Wilson 95 % intervals, completion rate and terminal-reason counts. Genuine engine-terminal draws are distinguished from timeouts, action caps, rejections, AI-driver faults and inference errors. A run with no completed games has an undefined win rate, not a zero score. Reporting the completion denominator alongside win rate is necessary because dropping difficult failed games can bias apparent strength. Decision traces record latency; aggregate p50/p95 includes attempted decisions in the retained game trace, including explicit inference failures. Candidate coverage and the number of decisions requiring a shortlist are also reported. Forced choices and warm/cold model starts must be separated in the eventual performance study.

**What remains controlled experimentally.** The pilot uses a fixed explicit deck and interleaves objective arms over the same picker-seed schedule. Replication across held-out deck archetypes, engine/AI seeds and seat assignments is planned after interface validation; we do not tune and report on the same holdout. The present chat bridge shares Tev1’s card-aware filtered projection, but offers all actions and permits explicitly traced random fallback; Tev1 uses a shortlist and surfaces inference errors. This is a *deployment-stack* comparison until context, shortlist and fallback accounting are matched. Shortlists of 8, 16 and 24 options, controlled option permutations and a size-matched general-model comparator are proposed ablations, not completed measurements. Benchmark outputs record model identifiers and configuration; runtime versions, quantisation, hardware and warm-up state must accompany published timing results.

**Data freeze and Commander qualification.** We maintain separate, versioned inputs: Scryfall oracle records and rulings for immutable card facts and graph construction, and the pinned engine’s compiled definitions for executable mechanics. Updating one does not update the other or retrain a policy. Evaluation must record source dates, checksums, deck payloads, engine/policy revisions and the support status of all study cards; graph population is validated against that same frozen snapshot.

The native pod wrapper can configure four Commander decks with one Python-controlled host and three native AI opponents. This is not a four-policy zoo tournament. Terminal pod qualification, elimination handling, transport limits and seat/deck rotation are separate gates; a turn-capped development smoke is never counted as a completed pod or draw. Multiplayer beliefs and diplomacy are not established by running a single-opponent latent planner in a four-player room.

**Publication sequence and statistical scope.** The executable first stage is a bounded feasibility pilot of runtime-qualified random, heuristic and objective stacks. It checks completion and audits action semantics before adding more arms. Decision models, matched chat rerankers, KG induction and learned JEPA/dreaming enter strength evaluation only after their interface, provenance and model-calibration gates pass. The standalone ontology resource can be evaluated independently; an early agents paper must restrict its empirical claims rather than imply that the whole proposed architecture has been measured.

**Current feasibility evidence and interface repairs.** After the interface repairs, a bounded native round-robin across burn, green creature aggro and blue-white control completed all 18 scheduled games: nine ordered deck matchups with one random and one heuristic episode each. Random won 2/9 and heuristic 3/9; all episodes had rules-terminal outcomes, with no incomplete cells and unchanged frozen input hashes. Four additional source-attributed historical deck mirrors also completed. These checks establish bounded execution across the selected fixtures, not comprehensive mechanics qualification or a reliable playing-strength difference.

**Replicated policy-stack execution.** A separate bounded campaign attempts twelve implemented conditions on a Modern mono-red burn mirror against `phase-ai` at `VeryEasy`, with three scheduled games per condition, picker seeds 7–9, a 1,500-action cap, a 115-second cooperative game deadline and a shared 360-second process deadline per condition. The campaign uses the pinned protocol-106 runtime and an owned private server; source, checkpoint and deck hashes and local model snapshots remain unchanged. `Tev1` uses the isolated same-weight 8,192-context alias. Chat output is bounded at 16 tokens with thinking disabled; this campaign precedes the full-action/card-aware chat prompt repair described below. Table 7 reports all attempts, including failures.

Gemma’s repeated subdecisions motivate a semantic prompt repair: listing only action types made distinct `ChooseOption` alternatives, including `Cancel` and `Cast`, indistinguishable. The revised prompt includes complete original action JSON and the shared perspective-safe card-aware projection. A separately frozen three-game repair check completes all three Gemma games, with 168 total decisions, zero wins and a 116.4-second total duration, under the same per-game and process limits. The activation audit identifies 30 successful model choices, 137 forced choices, and one request timeout invokes the explicitly traced fallback. We do not treat the completed games as proof of error-free inference. The repair is reported separately rather than replacing the earlier timeout or mixing prompt versions. Native randomness remains uncontrolled, so this is evidence of execution, not a causal estimate of prompt quality.

The other full-context chat selectors time out in a separate 165-second cooperative-budget check while turns continue to advance. With a declared 585-second cooperative and 600-second process limit, Llama and Qwen each complete one terminal loss in 194.6 and 48.1 seconds respectively. Their activation audits record 18 and six successful model choices, 140 and 133 forced choices, and six and one invalid-response fallbacks. These checks are not pooled with the shorter-budget campaign, and do not establish unaided LLM performance or reliable reply formatting.

The explicit model checkpoint was not trained on the native evaluation distribution, and permissive parameter loading requires a loaded-key audit before trained-model claims. No

| Condition                    | Completed / scheduled | Wins | Qualification boundary                            |
|------------------------------|-----------------------|------|---------------------------------------------------|
| Wire random                  | 3/3                   | 0    | Uniform offered-action baseline                   |
| Wire heuristic               | 3/3                   | 0    | Action-type tiers only                            |
| Chat: Llama 3.2 1B           | 3/3                   | 1    | 57 model choices; one invalid-response fallback   |
| Chat: Qwen2.5-Coder 1.5B     | 3/3                   | 0    | 53 model choices; no fallback                     |
| Chat: Gemma4 E2B             | 2/3                   | 0    | One timeout; 604 model choices across attempts    |
| Tev1 0.8B, 8K context        | 3/3                   | 0    | 39 model choices; no input cropping               |
| World model: direct          | 3/3                   | 0    | Checkpoint loading is not native learning         |
| World model: recurrent dream | 3/3                   | 0    | Two rollouts, depth three; not JEPA-head planning |
| LLM-model fusion             | 3/3                   | 1    | No graph supplied; activation unaudited           |
| KL control                   | 3/3                   | 1    | Analytic forecasts and horizon planning           |
| EFE information proxy        | 3/3                   | 0    | One-step proxy; budget differs from KL            |
| EFE ambiguity proxy          | 3/3                   | 1    | One-step proxy; budget differs from KL            |

Table 7: Replicated native feasibility: 35 of 36 scheduled games reach rules-terminal outcomes, with four wins and 31 losses. The remaining timeout is not a draw. These three-game cells do not satisfy the calibration gate or establish a policy ranking, a language-model contribution, or learning improvement.

graph-memory intervention is active in this campaign. The incomplete attempt remains visible rather than being removed from the denominator or replaced by easier fallback episodes. Thus these measurements demonstrate which stacks execute within the declared limits, not the effects specified by RQ2–RQ5.

The analytic control model distinguishes cancellation from pending-cast completion and interprets simple fixed-damage targets. These approximations do not establish card-/mode-exact dynamics or a causal objective comparison. Long Commander pods and live ontology-grounded decisions remain unqualified. A read-only ontology snapshot supports resource queries independently of gameplay; resource validation is not evidence that a playing policy used graph relations.

After pilot diagnostics, we predeclare comparisons and freeze a fresh holdout, including a fixed failure policy and sample size. Games are the analysis unit; the four correlated seats of a pod and its many actions are not independent replicates. We report stratum-specific effects and intervals, apply multiplicity control to confirmatory comparison families, and retain incomplete and unstarted cells separately. A 100-game binomial cell has approximately ten-percentage-point uncertainty near 50%; it cannot resolve a five-point advantage. An independent two-arm comparison near 50% needs roughly 1570 games per arm for 80% power at two-sided 5% significance to detect that effect, before additional dependence or multiplicity. Pilot durations therefore inform both a power analysis and an explicit compute budget, not an automatic multi-day run.

### 6.3 Controlled objective and ontology interventions

**Objective ablation for the control formulation.** The claim of Section 3.3 — that belief-space KL control is the right objective and the two conventional EFE factorisations are not

— is an empirical claim in our domain, and we state the experiment that tests it rather than asserting the conclusion. All three objectives are exposed by one picker interface. However, the current KL arm performs horizon planning while the legacy EFE arms use one-step scoring. The shared belief, transition model and preference potential do not remove this planning-budget confound. The implemented interleaved episode sweep is consequently a policy-stack pilot; an objective-only ablation requires a common continuation set, horizon and computational budget. Picker seeds alone also do not seed the native engine or opponent AI.

Three measurements are planned against **phase-ai** at fixed difficulty:

1. **Win rate.** If the analysis transfers, **kl** should dominate both surrogates.
2. **An irrelevant-information probe.** Construct matched positions with a learnable but strategically irrelevant observation, controlled action/resource costs and unchanged attainable outcome values. Compare selections with and without the epistemic bonus; include irreducible random noise as a negative control. Raw cantrip counts or an unchanged immediate action do not suffice, because draw quality, material and later contingencies may still matter.
3. **An ambiguity-aversion probe.** Define the observation channel explicitly and separate conditional observation entropy, belief uncertainty and reward variance. Compare matched tactical choices with equal deviation costs and evaluated outcome preferences, varying only the ambiguity term. Declined attacks or open mana are descriptive statistics, not proof of a pathological entropy penalty.

The existing legacy arms are proxy baselines, not full theoretical observation-channel implementations. A controlled result must qualify their correspondence to the equations before generalising to Active Inference. No separation would also be informative, but could reflect limited horizons, model errors or inadequate probes rather than establishing equivalence of the objectives.

**Ontology-conditioned evaluation.** The ontology is validated structurally and against its own competency questions (Neubürger and Neubürger, 2026), but the *agent-facing* value of the symbolic layer is not yet measured. Two follow-ups are queued. First, a reasoning benchmark generated from the ontology itself by backward-chaining proof search in the manner of Van Schependorn et al. (2026), which gives controlled multi-hop depth and proof-based negatives for free. Second, submission of the generated artefact to a public competency-question-driven ontology benchmark (Ontology Engineering Group, Universidad Politécnica de Madrid, 2026) as an additional baseline, since our construction combines author-directed, LLM-assisted conceptual modelling with deterministic normative vocabulary extraction rather than an autonomous generation pipeline.

We deliberately do not claim “state of the art” numbers: we know of no comparable public baseline that plays full-Comprehensive-Rules MTG, and the contribution of this report is the system itself, plus the ontology and the released self-play infrastructure.

## 7 Discussion and Limitations

**Scope.** The engine is **phase-rs** and therefore covers the Comprehensive Rules to whatever depth upstream does; our own exposure is the adapter, which currently flattens the new viewer-interaction schema into the same opaque choice list as ordinary actions and so cannot yet reason about *what kind* of choice it is making. The ontology v2.0 covers the core game structure, the ability taxonomy, the strategic layer and the agent decision module, but does not yet encode every keyword from every set, and we do not extend it automatically per Standard rotation.

**Ontology limitations.** Three are worth stating plainly. (i) The import-resolved schema has passed a local Hermit consistency/satisfiability run, but independent DOLCE modelling review



and complete profile conformance remain unqualified: the unmodified external documents have OWLAPI profile violations, and a schema-plus-vocabulary run exceeded 600 seconds. Focused positive/negative object/event probes do not certify a populated graph; CR 205.3 is checked procedurally. A read-only Fuseki snapshot reproduces 14 SPARQL checks, not agent grounding or implicit reasoning. (ii) Card designs now instantiate the type, subtype and quality/region patterns, but the role/situation pattern has no population path yet — roles are produced only by gameplay, and the trace-to-graph writer for combat situations is not implemented. (iii) Coverage is measured against our own competency questions, which is a weak test: a question set authored alongside an ontology will tend to be answerable by it. Extrinsic evaluation against an external benchmark is the honest check and is queued in Section 8.

**Representation and computation.** The present representation has 256 latent dimensions, and recurrent dynamics retain a separate history state. Larger models, deeper rollouts and more samples change inference cost; whether they improve playing strength must be measured. Planning accuracy cannot be inferred from model size or one-step training loss.

**The control formulation is adopted, not re-derived.** We use the belief-space objective motivated by [Kaufmann \(2026\)](#); the contribution is its proposed domain transfer and controlled evaluation, not a new equivalence theorem. Native KL rollouts presently use an analytic approximation. A learned latent predictor is a proposed replacement, not a JEPA decoder of complete game states. The tree planner’s default continuation set is narrow, while the current sampled planner uses open-loop continuations. Neither presently establishes the full contingent planning needed to demonstrate dual-control information value.

**LLM grounding.** Selectors operate over engine-provided options rather than inventing action objects. This constrains the executable output, but does not guarantee strategically grounded reasoning. Invalid text, unavailable inference and incomplete candidate projections remain explicit failure modes. Using model preferences as a planner prior is distinct from direct option selection.

**Decision-model limitations.** Tev1’s decision format fits the agent interface, but its training recipe is not MTG-specific and it does not simulate consequences. A plausible choice distribution may reflect textual or option-order bias rather than tactical understanding. Restricting candidates can hide the best move, and action coverage is not an optimal-action oracle. Confidence concentration is not calibration. The model can therefore complement, but does not substantiate or replace, the learned-dynamics and control claims of this paper.

## 8 Future Work

The controlled studies in Section 6.1 motivate the following extensions:

1. **Controlled decision-model evaluation.** Complete the matched-context, matched-shortlist and size-matched comparisons of Section 6.2; instrument chat fallback, seed the native engine/AI and rotate seats. Then evaluate Tev1 as a critic over world-model candidates separately from its direct-selector baseline.
2. **Native learning and component interventions.** Train and calibrate latent dynamics, reward and policy components on frozen native trajectories with independent training seeds. Evaluate the weight/graph factorial and leave-component-out fusion under activation audits, before attributing an effect to JEPA, a graph or a language model.

3. **Deck-diversity generalisation.** A larger study should cover distinct strategic demands: tempo, midrange, control, ramp, graveyard recursion, artifact synergy, prison and multi-card combo, rather than mainly variations on burn. Include multiple independently constructed lists per family and hold out complete construction families; audit shared cards and combo packages to prevent near-duplicate leakage. Measure within-family piloting, unseen-list adaptation and unseen-playstyle transfer separately, with equal exposure across producers and per-archetype outcomes rather than only a pooled win rate. Cross-format and Commander transfer should be separate experiments because legality, starting life, deck size and multiplayer incentives also change. The present seven-list pool contains related burn variants and cannot substantiate this broad generalisation claim.
4. **Strategy-aware opening decisions.** Compare archetype-sensitive keep/bottom policies under matched information and mulligan rules; evaluate learned forecasts only after their opening-hand calibration is established.
5. **Commander politics.** Threat assessment across  $N > 2$  opponents, including a multi-agent extension of the active-inference objective, and the Commander variant of the mulligan (free first mulligan).
6. **Ontology evolution and ABox migration.** Migrate the Scryfall importer onto the v2.0 role/situation and quality/region patterns so the expressive gain reaches the populated graph, then wire the framework into a human-in-the-loop ontology-evolution toolchain (Data Science Lab, Fachhochschule Südwestfalen, n.d.; jiayi et al., 2026) so the schema is updated semi-automatically per Standard rotation, with humans adjudicating debated axioms. The retrieval-scoped extension strategy of Lippolis et al. (2026) — retrieve the relevant ontology fragment per competency question rather than stuffing the whole TBox into context — is the approach we intend to follow.
7. **Promotion of induced evidence.** Define and evaluate the gate by which an `mtg:InducedSynergy` with sufficient support may be promoted to a curated `synergisesWith` edge, including how to represent and surface disagreement between runs (Ehrenmüller et al., 2026). This is the point at which the curated/induced boundary becomes a policy question rather than a schema question.
8. **Structured interaction handling.** Consume `phase-rs`’s viewer-interaction schema natively instead of flattening it, so the agent can distinguish a targeting decision from a mode choice from a damage assignment — each of which has a different shape of belief dependence.

## 9 Conclusion

We have presented a research framework for combining native rules execution, structured knowledge and learned planning in Magic: The Gathering. The system implements candidate components and a completion-aware native evaluation interface, but does not yet demonstrate the complete ontology/KG/JEPA learning architecture end-to-end. Current evidence consists of resource validation and bounded native execution measurements. The self-contained study design separates policy comparison, parametric learning, graph memory, their interaction and cross-agent transfer. Controlled strength comparisons, live graph grounding and multiplayer qualification remain prerequisites for empirical learning claims.

**Reproducibility statement.** Code, the OWL ontology, sample decklists, training scripts and this report are available at <https://github.com/cuteredpwnda/opposition-agents-p-laying-mtg>. Project-authored materials use the repository’s MIT license; third-party decks and dependencies retain their accompanying licenses and attribution. Experimental manifests identify source revisions, deck and checkpoint hashes, configuration and retained failure records.

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